

ECON 101—Introduction to Microeconomics

Spring 2026

MIDTERM I

February 19, 2026

General Instructions

- Answer all questions. **All answers should be written in the answer sheet.** You can use as many scratch paper for your preliminary calculations as well as the exam booklet, but that work will not be considered. I will only consider the answers written in the answer sheet.
- At the end of the exam, you can take this booklet with you.
- The exam has a duration of **60 minutes**.
- You can request a bathroom break, but only one person can be out at any given time.
- On your desk, you are allowed to have your ID, writing instruments, a ruler, and a simple calculator (no graphing calculators). You may also have a simple watch if needed (smartwatches are not allowed).
- Please place all other personal items, including smart devices, to the side of the room.
- If you complete the exam early, you may leave the room. but only within the first 45 minutes. Otherwise, please remain in the room quietly until the exam concludes.
- Crying is allowed, but please do so quietly to avoid disturbing others.
- Do not use the exam paper to wipe away tears.
- Good luck and may your choice of inputs maximize your profits in this exam!

Academic Dishonesty Policy and Guidelines at the End of the Exam

- Any form of cheating will be met with the **harshest consequences available for the level of the offense**. There will be **no sympathy** and **no leniency**.
- At the end of the exam, I will announce that the time is up. You **must follow these instructions**:
 - Put all your writing instruments down and close your exam booklet.
 - Keep your hands away from any writing instruments and your exam booklet.
 - Remain quiet and orderly. You are not free to leave until instructed to do so.
 - Any deviation from these rules will raise suspicion of academic dishonesty, and **you will be penalized accordingly**.

How to fill the bubbles and total points.

- Please use the answer sheet to record your answers. Only the answers in the answer sheet will be considered!
- Please use **only the provided pencil** to fill the bubbles! If you make a mistake be sure to erase your selection properly with the provided eraser!
- Examples:
 - Example: Correct answer. Q.1.  (B) (C) (D)
 - Example: Incorrect way to mark an answer. Q.2.  (B) (C) (D)
 - Example: Answer marked wrong because two bubbles are filled. Q.3.  (C) (D)
- The exam has 25 questions worth 4 points each for a total of 100 points.

Please record all answers in the answer sheet. Each question is worth 5 points

1. A consumer's willingness to pay for successive cups of coffee during a given week is:

Cup number	1	2	3	4	5
Willingness to pay	50	40	30	20	5

The price of coffee is $P = 25$ per cup.

The coffee shop introduces the following promotion: if a consumer purchases 4 cups during the week, they are entitled to receive a 5th cup at zero price.

How many cups will the consumer choose to consume during the week? (*Think in terms of cost-benefit analysis.*)

- (a) The consumer is indifferent between consuming 3 cups and 5 cups.
- (b) 3 cups.
- (c) 5 cups.
- (d) 4 cups.

The following information is used in questions 2–8. In the economy of NUBurgh, coffee consumption is divided across three distinct student archetypes:

- Type 1: “The PhD Energy Vampire” — survives exclusively on coffee.
- Type 2: “The Responsible Senior” — drinks coffee but claims it is only for productivity.
- Type 3: “The First-Year Idealist” — drinks coffee only during midterms.

Each type has the following inverse demand for coffee:

$$P_1(Q_1) = 30 - Q_1, \quad P_2(Q_2) = 20 - Q_2, \quad P_3(Q_3) = 10 - Q_3,$$

where $Q_i = 0$ whenever the implied quantity would be negative.

Coffee is supplied competitively by sleep-deprived café owners, with market supply:

$$Q^s(P) = -5 + P,$$

and supply is zero whenever the implied quantity would be negative.

2. The aggregate demand function for coffee in NUBurgh is:

- (a) $Q^d(P) = \begin{cases} 60 - 3P, & P \leq 20, \\ 50 - 2P, & 20 < P \leq 30, \\ 0, & P > 30. \end{cases}$
- (b) $Q^d(P) = 60 - 3P$ for all $P \leq 30$, and $Q^d(P) = 0$ for $P > 30$.
- (c) $Q^d(P) = \begin{cases} 60 - 3P, & P \leq 10, \\ 50 - 2P, & 10 < P \leq 20, \\ 30 - P, & 20 < P \leq 30, \\ 0, & P > 30. \end{cases}$
- (d) $Q^d(P) = \begin{cases} 30 - P, & P \leq 10, \\ 50 - 2P, & 10 < P \leq 20, \\ 60 - 3P, & 20 < P \leq 30, \\ 0, & P > 30. \end{cases}$

3. The competitive equilibrium price and quantity in the NUburch coffee market are:

- (a) $P^* = \frac{55}{2}$ and $Q^* = \frac{45}{2}$.
- (b) $P^* = \frac{55}{3}$ and $Q^* = \frac{40}{3}$.
- (c) $P^* = 20$ and $Q^* = 15$.
- (d) $P^* = 15$ and $Q^* = 10$.

4. At the competitive equilibrium, the consumer surplus in the NUburch coffee market is:

- (a) $\frac{125}{3}$
- (b) $\frac{800}{9}$
- (c) $\frac{625}{9}$
- (d) $\frac{200}{3}$

5. At the competitive equilibrium, the producer surplus in the NUburch coffee market is:

- (a) $\frac{625}{9}$
- (b) $\frac{400}{9}$
- (c) $\frac{800}{9}$
- (d) $\frac{200}{3}$

6. Suppose the mayor of NUburch introduces a per-unit tax of $t = 10$ on coffee, collected from sellers. Let p^d denote the price paid by consumers and p^s the price received by producers, so that $p^d = p^s + 10$. The new competitive equilibrium prices and quantity are:

- (a) $p^{d*} = \frac{45}{2}$, $p^{s*} = \frac{25}{2}$, and $Q^* = \frac{15}{2}$.
- (b) $p^{d*} = \frac{35}{2}$, $p^{s*} = \frac{15}{2}$, and $Q^* = \frac{25}{2}$.
- (c) $p^{d*} = 20$, $p^{s*} = 10$, and $Q^* = 10$.
- (d) $p^{d*} = \frac{55}{3}$, $p^{s*} = \frac{25}{3}$, and $Q^* = \frac{40}{3}$.

7. Under the tax, the deadweight loss is:

- (a) $\frac{800}{9}$
- (b) $\frac{325}{12}$
- (c) $\frac{225}{8}$
- (d) 75

8. Under the tax, which consumer types purchase coffee in equilibrium?

- (a) Types 1 and 2 only.
- (b) Types 2 and 3 only.
- (c) Type 1 only.
- (d) Types 1, 2, and 3.

9. Medina tells Akmaral that the price of cinema tickets increased from 1,450 to 2,900 tenge, and she now goes to the cinema half as often as before.
What happened to Medina's total spending on cinema tickets?
- (a) It increased.
 - (b) There is not enough information to determine this.
 - (c) It decreased.
 - (d) It remained the same.

The following information is used for questions 10–11. NUburch is a small open economy. Coffee beans are traded internationally at the exogenous world price $p^w = 10$. NUburch takes this price as given, so under free trade the domestic consumer price equals the world price.

Domestic demand and domestic supply are:

$$Q^d(p) = 100 - 5p, \quad Q^s(p) = -20 + 4p,$$

with the convention that quantities are zero whenever the expression is negative.

The mayor introduces a per-unit subsidy of $s = 3$ to local coffee producers. Let p^d denote the price paid by consumers and p^s the price received by producers.

10. Under the subsidy, the domestic prices, domestic production, and imports are:

- (a) $p^d = 10$, $p^s = 7$, $Q^s = 8$, and imports = 42.
- (b) $p^d = 10$, $p^s = 10$, $Q^s = 20$, and imports = 30.
- (c) $p^d = 13$, $p^s = 13$, $Q^s = 32$, and imports = 0.
- (d) $p^d = 10$, $p^s = 13$, $Q^s = 32$, and imports = 18.

11. The deadweight loss generated by the producer subsidy is:

- (a) 96
- (b) 9
- (c) 18
- (d) 0

12. Sultan's income increases from 2000 to 2500, and his consumption of Coke increases from 30 to 50 units, holding prices constant.

Based on this information, which of the following statements must be true?

- (a) Coke is an inferior good.
- (b) The demand for Coke is perfectly elastic.
- (c) The price elasticity of demand for Coke is greater than one.
- (d) Coke is a normal good.

13. The market for chairs is initially in equilibrium. New scientific evidence suggests that chairs are better for spinal health than sofas, increasing consumers' willingness to buy chairs. At the same time, producers expand production capacity for chairs.

What happens to the equilibrium price and quantity in the market for chairs?

- (a) Both price and quantity increase.
- (b) Both price and quantity decrease.
- (c) Price increases, but the effect on quantity is ambiguous.
- (d) Quantity increases, but the effect on price is ambiguous.

Read carefully the following text and answer questions 14–18. The economy of NUburch produces the following five consumption goods: Coffee (Co), Energy drinks (En), Private tutoring (T), Clubbing nights (Cl), and Instant noodles (N).

The following conditions hold on the income elasticities of demand:

$$\epsilon_I^N < 0 < \epsilon_I^{Co} < \epsilon_I^{En} < 1 < \epsilon_I^T < \epsilon_I^{Cl}.$$

Furthermore, we know that the following relationships are true:

- *Co* and *En* are substitutes.
- *Co* and *T* are complementary.
- *En* and *T* are complementary.
- *T* and *Cl* are substitutes.
- *Co* and *Cl* are complementary.
- *En* and *Cl* are complementary.
- *N* and *T* are complementary.
- *N* and *Cl* are substitutes.
- *Co* and *N* are complementary.

The supply curves of all the goods have positive slope.

Answer the following **five** questions, 14–18, in strict order, only taking into account the effects of the previous questions to your answer.

Suppose that there is a technological change in the production of coffee.

14. In the coffee (*Co*) market:

- (a) The supply curve will shift to the right and the quantity demanded will increase.
- (b) Only the demand curve will shift, neither the supply curve nor the equilibrium quantity will be affected.
- (c) The demand and supply curves will shift.
- (d) The demand curve will shift to the right and the quantity offered will increase.

15. In the energy drink (*En*) market (consider exclusively the change in price in the coffee market)

- (a) The price and the equilibrium quantity will decrease and the producer surplus will decrease.
- (b) The price will decrease and the equilibrium quantity will increase and the producer surplus will increase.
- (c) The demand and supply curves will both shift to the left.
- (d) The price and the equilibrium quantity will increase and the producer surplus will increase.

16. In the private tutoring (*T*) market (consider exclusively the change in price in the coffee and energy drink markets)

- (a) The demand curve will shift to the right and the equilibrium quantity will increase.
- (b) The demand and supply curves will both shift to the right.
- (c) Only the supply curve will shift to the right and the equilibrium quantity will increase.
- (d) The demand curve will shift to the left and the equilibrium quantity will decrease.

17. In the clubbing (*Cl*) market (consider exclusively the change in price in the coffee, energy drink, and tutoring markets)

- (a) Only the supply curve will shift to the right, and the equilibrium quantity will increase.
- (b) The demand curve will shift to the left and the equilibrium quantity will decrease.
- (c) The demand curve will shift to the right and the equilibrium quantity will increase.
- (d) The demand curve will shift to the right but the equilibrium quantity will decrease.

18. If the income of the population increases (consider exclusively the change in income)
- The demand curve for all goods will shift to the right.
 - The demand curve of all goods except instant noodles will shift to the right.
 - The demand curve for coffee and energy drinks will shift to the left.
 - The equilibrium quantity in all markets will increase.
19. Five individuals decide to open a bakery. Each has 60 minutes available and can allocate their time between baking cakes and cookies. Productivity is as follows:
- Nurai and Malika can bake 1 cake in 60 minutes or 1 cookie in 30 minutes.
 - Medina can bake 1 cake in 30 minutes or 1 cookie in 15 minutes.
 - Akmaral and Sultan can bake 1 cake in 45 minutes or 1 cookie in 20 minutes.

Cookies are measured on the horizontal axis and cakes on the vertical axis. Consider the *joint* PPF of the bakery (assuming full specialization according to comparative advantage and linear production within each individual).

How many kinks does the joint PPF have?

- 5
- 4
- 1
- 0

Read carefully the following information, which is used in questions 20 to 23. Anel, Daryn, Malika, Nuray, and Tamerlan enthusiastically signed up for what was advertised as an immersive *Game of Thrones*-style live-action experience, complete with political intrigue, strategic alliances, and “authentic medieval realism.”

Upon arrival, they discovered that the “realism” consisted primarily of being dropped into an isolated field with basic tools and a strongly worded disclaimer. With no catering, no stunt coordinators, and no dragons (disappointingly), they must now divide their time between foraging food (measured in kilograms) and constructing defensive fortifications (measured in square meters). Each of them can work up to 10 hours per day. With one hour of work, they can produce the following:

	Food (kg)	Fortifications (m ²)
Anel	8	2
Daryn	4	6
Malika	7	3
Nuray	2	9
Tamerlan	5	4

20. Based on the productivity table above, who has the comparative advantage in foraging food, and who has the comparative advantage in building fortifications?
- Tamerlan in food; Daryn in fortifications.
 - Anel in food; Nuray in fortifications.
 - Anel in food; Tamerlan in fortifications.
 - Malika in food; Daryn in fortifications.

21. The camp organizers open a trading post offering the following terms of trade: 2 units of fortifications (D) in exchange for 3 units of food (F). Assuming each individual specializes according to comparative advantage at these terms of trade, at which point will the group specialize, and who produces what?
- The group specializes at (240, 90); Anel produces food; the others produce fortifications.
 - The group specializes at (200, 150); Anel, Malika, and Tamerlan produce food; Daryn and Nuray produce fortifications.
 - The group specializes at (80, 220); only Anel produces food; all others produce fortifications.
 - The group specializes at (150, 190); Anel and Malika produce food; Daryn, Nuray, and Tamerlan produce fortifications.
22. Suppose the trading post offers fixed terms of trade between food (F) and fortifications (D). For which of the following offers will the group fully specialize in the production of fortifications?
- 2 units of F in exchange for 1 unit of D .
 - 8 units of F in exchange for 2.4 units of D .
 - 8 units of F in exchange for 3 units of D .
 - 8 units of F in exchange for less than 2 units of D .
23. For which of the following conditions will exactly three individuals specialize in food production and two specialize in fortifications?
- 5 units of F exchange for less than 4 units of D , and 2 units of F exchange for more than 3 units of D .
 - 5 units of F exchange for more than 4 units of D , and 2 units of F exchange for less than 3 units of D .
 - 5 units of F exchange for less than 4 units of D , and 2 units of F exchange for less than 3 units of D .
 - 5 units of F exchange for more than 4 units of D , and 2 units of F exchange for more than 3 units of D .
24. In the town of NUtopia, the townspeople like to consume lattes. There are two types of consumers: regular coffee drinkers (regulars R) and coffee snobs (snobs S). Assume both demand curves are downward sloping. Suppose that the market equilibrium happens at the same price and quantity in the regulars' market and in the snobs' market. At the market equilibrium, the price elasticity of demand for regulars is $\epsilon_{D,R}^P = -2$ and the price elasticity of demand for snobs is $\epsilon_{D,S}^P = -1.2$. Assume that the supply of lattes in NUtopia is perfectly elastic. In a bold display of what can only be described as visionary fiscal creativity, the NUtopian government introduces a per-unit tax of $\$t$ on lattes (with $t > 0$), confident that the townspeople will appreciate the policy in due time. We can be sure that: the decrease in consumer surplus for coffee snobs will be _____ than the decrease in consumer surplus for regulars. The deadweight loss will be greater in the market of _____.
- greater; regulars.
 - lower; regulars.
 - greater; snobs.
 - lower; snobs.
25. The University of Quantifiable Prestige allocates a substantial portion of its annual budget to marketing campaigns aimed at improving its position in international rankings, rather than allocating those funds elsewhere within the institution. The opportunity cost of the funds spent on marketing is _____.
- Zero, since improving rankings may generate long-term benefits.
 - The value of the next best alternative use of those funds, such as hiring more faculty or funding additional research grants.
 - The uncertainty about whether the rankings will improve.
 - The accounting cost of the marketing campaign.